# 99.500 Applied Mathematics for Engineering

Computer-based Test

April 18, 2026

This 120-minute open-book assessment allows internet access but no communication. Total 100 marks will be rescaled to 20% of your final grade. Show all working and attempt all parts of each question. Partial credit will be given where appropriate.

**Question 1 (55 Marks)**: PDE Problem — Cooking the Perfect Steak
Mr. AME is a consultant for a high-tech kitchenware company developing a smart pan designed to cook steak perfectly every time. One of the key objectives is to determine how long it takes for the center of the steak — the coldest point — to reach an appetizing temperature. To achieve this, he has formulated a model for this cooking process in which the temperature distribution $T(z,t)$ in the steak is governed by the 1D transient heat conduction equation:

$$\frac{\partial T}{\partial t} = \kappa \frac{\partial^2 T}{\partial z^2} \text{ with } \begin{cases} T(z,0) = T_0 \\ T(0,t) = T_p \\ -\alpha \left. \frac{\partial T}{\partial z} \right| = \gamma \left[ T(\beta,t) - \delta T_p \right] \end{cases} \tag{1}$$

where $\kappa$ is the thermal diffusivity, $\alpha$ is the material conductivity, $\beta$ is the steak thickness, $T_0$ is the uniform initial temperature of the steak, $T_p$ is the temperature at pan side of the steak in direct contact with the heat source, and $\gamma \left[ T(\beta,t) - \delta T_p \right]$ represents that the top surface loses or gains heat depending on the surrounding temperature which is modeled as proportional to the pan temperature $T_p$ with proportional coefficient $\delta$. Assume the physical parameters have the following values:

$$\kappa = 0.1, \ \alpha = 0.05, \ \beta = 2, \ \gamma = 3, \ \delta = 0.9, \ T_0 = 5$$

The only variable parameter is the pan-side cooking temperature, $T_p$ , which the chef can adjust.

Numerically solve the PDE system above with linspace(0,2,50) for $x$ & suitable domain and mesh for $t$. Study how different values of $T_p$ affect the cooking time. Specifically, determine the time it takes for the coldest point of the steak (the center, or lowest temperature across the domain) to reach a target internal temperature $T_i = 10$.

| $T_p$ | Time $t$ when $T_i = 10$ |
|---|---|
| 15 | |
| 25 | |
| 35 | |
| 45 | |

(a) [10 Marks] Fill in the table above with the estimated cooking time for each $T_p$. Approximations are acceptable.

(b) [20 Marks] On the same graph, plot the temperature profiles at the moments when the center reaches $T_i = 10$, for all $T_p$.

(c) [10 Marks] Identify and briefly explain your top 3 key findings from the simulation results.

(d) [5 Marks] Qualitatively discuss what happens if the chef forgets to turn off the heat — i.e., continues heating beyond the target temperature.

(e) [10 Marks] Provide your complete numerical code for the case $T_p = 45$.

**Question 2 (45 Marks)**: ODE Problem — Blasius Boundary Layer

In many fluid dynamics applications, a third-order nonlinear ordinary differential equation arises from the transformation of partial differential equations using similarity variables. Consider the following boundary value problem (BVP) involving a function $f(\eta)$:

$$2f''' + f''f = 0 \text{ with } \begin{cases} f(0) = 0 \\ f'(0) = 0 \\ f'(\infty) = 1 \end{cases} \tag{2}$$

This equation represents the boundary layer flow over a flat plate. As computational tools cannot evaluate expressions at $\infty$, the boundary condition at infinity can be approximated by imposing it at a sufficiently large value of $x$, beyond which the solution changes negligibly.

(a) [25 Marks] Using a numerical method of your choice to solve the above BVP for four different domain lengths $\eta \in [0, L]$, where $L = 5$, 20, 35, 50 with a consistent number of mesh points per interval (100 points per 1 unit of $\eta$), and experiment with two different initial guesses for $f(\eta)$ ($f_0(\eta) = 0$ and $f_0(\eta) = \eta$). For each value of $L$, present your numerical solution using plots of $f(\eta)$, and include any error information if applicable.

(b) [10 Marks] Present and discuss top 3 important findings, based on your observations and resutls in part (a).

(c) [10 Marks] Provide your complete numerical code for the case $L = 50$ and $f_0(\eta) = \eta$.